Video 05 | Envelopes and LFOs

Massive PSXL instructional series - video script

Runtime estimate	Narration length	Scenes	Format
about 5:27 minutes	769 words at 150 words per minute	13	Screen capture of the plug-in with voice-over; dark theme, interface size 125 percent

Learning goals

- Edit the amp envelope and understand Delay, Attack, Hold, Decay, Sustain, Release and OneShot.
- Use Mod 2 to Mod 9 as envelopes or LFOs and assign them by dragging chips.
- Edit amounts and understand voice versus global modulation.

> **NOTE.** Accuracy rule for the narrator and any generated video: use the on-screen names exactly as written in the action column. After **Load Init (Blank)** the synth is silent until routing is connected. Do not compare Massive PSXL to other products.

Scene table

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
1	Init (Blank), Osc 1 > Filter 1 > Out, Osc 1 Pos 0.66, Filter 1 Cutoff 700, Reso 0.3. Open Amp Env tab.	Envelopes shape how a sound changes over a note. We start from a saw through a low-pass filter, and open the Amp Env tab. Everything in this video happens with this one simple patch, so you can follow along exactly.	Show the envelope display large.
2	Drag the three handles on the display; then use the knobs Delay, Attack, Hold, Decay, Sustain, Release, Env Level.	The graph has three handles. The first sets Attack, the middle sets Decay across and Sustain up and down, the last sets Release. The knobs also give you Delay, Hold and Env Level. The dot shows where your latest note is in the envelope. Delay waits before the attack begins, and Hold keeps the envelope at its peak for a moment before it decays. Env Level scales the whole envelope, which is useful when the envelope is used as a modulator.	Label the stages on the graph: Delay, Attack, Hold, Decay, Sustain, Release.
3	Preset the envelope to Attack 0.001, Decay 0.3, Sustain 0, Release 0.2; hold a key. Then switch OneShot on and tap a key briefly.	With sustain at zero, the note decays to silence even while you hold the key. Now switch on OneShot. A quick tap now plays the whole decay, because a one-shot envelope ignores key release. That is the secret to drums. Decay and release curves are exponential, so they sound natural for plucks and percussion. A new attack always starts from the current value, so retriggering never clicks.	Show a short tap with OneShot off (cut) versus on (full).
4	Open Mod 2 (Type Envelope): Attack 0.001, Decay 0.25, Sustain 0, Release 0.2. Drag the Mod 2 chip onto Filter 1 Cutoff.	Mod 2 to Mod 9 are slots that can be envelopes or LFOs. Mod 2 is an envelope. Drag the chip at the top left of the tab onto the Cutoff knob. A ring appears around the knob. That is the assignment. The chip at the top left of each modulation tab is the source. The knob you drop it on is the destination. The ring that appears around the knob shows how far the source can move it.	Animate the chip being dragged with the crosshair; zoom on the ring appearing.

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
5	Right-click Cutoff > Edit amount: Mod 2; drag the slider to +0.4, then -0.4, then back.	Right-click the knob and choose Edit amount. The slider goes from minus one to plus one. Positive opens the filter on the attack, negative closes it. Amounts are fractions of the knob's travel, so plus point four moves it forty percent of the way up. While you play, a white dot on the knob shows the live modulated value. It is a great way to understand what is actually happening to the sound.	Show the callout slider with value readout. Show the white dot on the knob moving with the envelope.
6	Mod 3: Type LFO, Shape Sine, Rate 5 Hz. Drag Mod 3 chip onto the Pitch knob in the Pitch panel; amount +0.003; Fade In 0.6.	Now an LFO. Mod 3 becomes an LFO by changing Type. Sine at five hertz, dragged onto the Pitch knob with a tiny amount, point zero zero three, is a gentle vibrato. Fade In brings it in after the note starts. A very small amount on the Pitch knob goes a long way, because the Pitch knob covers ninety-six semitones. Zero point zero zero three is only about a third of a semitone.	Zoom on the LFO display showing the sine and playhead.
7	Cycle through the seven LFO shapes with the Shape menu: Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random.	There are seven shapes: sine, triangle, saw up, saw down, square, sample and hold, and smooth random. Sample and hold jumps to a new random value each cycle; smooth random glides between them. Level controls how deep the LFO goes. With the level at half, the same amount produces half the movement, so you can use it as a depth control for the whole slot.	Show the shape in the LFO display for each choice.
8	Mod 4: LFO Square, Sync on, Division 1/8, assign to Filter 1 Cutoff amount +0.3; toggle Bipolar off and on.	Turn on Sync and the Rate knob becomes a Division menu, locked to your project tempo. Eighth notes on the cutoff gives rhythmic movement. Bipolar on swings above and below the knob; off, it only goes one way. If you are working in a DAW the synced LFO follows the project tempo. In the standalone app, where there is no host tempo, it assumes one hundred and twenty beats per minute.	Overlay a metronome clicking to show sync. Mention 120 BPM if there is no host tempo.
9	Mod 4 Phase knob, Level knob demo; mention each LFO is per note.	Phase sets where the LFO starts at each note. Level sets its depth. LFOs are per voice, so every note you play starts its own cycle. Because each note has its own LFO, chords can sound alive and slightly different on each note, unless the phase is set to start them all together.	Short animation of two overlapping notes with offset LFO phases.
10	Drag the Mod 3 chip onto the Reverb Mix knob in the Verb tab. Toast appears.	One rule to remember: voice modulators, envelopes, LFOs, performers, velocity, key track and random, cannot control the master effects. Try it and you get a message. Effects are shared by all notes, so for them use a macro, mod wheel or aftertouch. That is why macros, the mod wheel and aftertouch are the way to move effects. They are the same for every note, so they are meaningful for the shared effects at the end of the chain.	Show the toast text: 'Voice modulators can't control global effects. Use a macro, mod wheel or aftertouch.'
11	Drag the ModWhl chip (macro strip) onto Reverb Mix; move the on-screen mod wheel.	Drag the ModWhl chip instead and it works. Move the wheel on the on-screen keyboard and the reverb opens.	Show the keyboard drawer with the wheel moving.
12	Right-click Cutoff > Remove Mod 2; Undo; Redo.	To remove a modulation, right-click the knob and choose Remove. Undo and Redo cover assignments, so you can experiment freely. You can have up to two hundred fifty-six assignments, several sources on one knob, and one source on many knobs, so do not be afraid to layer modulation.	Show Undo and Redo buttons lighting.
13	End on the finished patch; roll end card.	Envelopes for the shape of a note, LFOs for repeating movement. In video six we meet macros, the controls that rename themselves.	End card: 'Next: 06 Macros and automatic naming'.

Narration only (plain script for text-to-speech or NotebookLM)

[Scene 1] Envelopes shape how a sound changes over a note. We start from a saw through a low-pass filter, and open the Amp Env tab. Everything in this video happens with this one simple patch, so you can follow along exactly.

[Scene 2] The graph has three handles. The first sets Attack, the middle sets Decay across and Sustain up and down, the last sets Release. The knobs also give you Delay, Hold and Env Level. The dot shows where your latest note is in the envelope. Delay waits before the attack begins, and Hold keeps the envelope at its peak for a moment before it decays. Env Level scales the whole envelope, which is useful when the envelope is used as a modulator.

[Scene 3] With sustain at zero, the note decays to silence even while you hold the key. Now switch on OneShot. A quick tap now plays the whole decay, because a one-shot envelope ignores key release. That is the secret to drums. Decay and release curves are exponential, so they sound natural for plucks and percussion. A new attack always starts from the current value, so retriggering never clicks.

[Scene 4] Mod 2 to Mod 9 are slots that can be envelopes or LFOs. Mod 2 is an envelope. Drag the chip at the top left of the tab onto the Cutoff knob. A ring appears around the knob. That is the assignment. The chip at the top left of each modulation tab is the source. The knob you drop it on is the destination. The ring that appears around the knob shows how far the source can move it.

[Scene 5] Right-click the knob and choose Edit amount. The slider goes from minus one to plus one. Positive opens the filter on the attack, negative closes it. Amounts are fractions of the knob's travel, so plus point four moves it forty percent of the way up. While you play, a white dot on the knob shows the live modulated value. It is a great way to understand what is actually happening to the sound.

[Scene 6] Now an LFO. Mod 3 becomes an LFO by changing Type. Sine at five hertz, dragged onto the Pitch knob with a tiny amount, point zero zero three, is a gentle vibrato. Fade In brings it in after the note starts. A very small amount on the Pitch knob goes a long way, because the Pitch knob covers ninety-six semitones. Zero point zero zero three is only about a third of a semitone.

[Scene 7] There are seven shapes: sine, triangle, saw up, saw down, square, sample and hold, and smooth random. Sample and hold jumps to a new random value each cycle; smooth random glides between them. Level controls how deep the LFO goes. With the level at half, the same amount produces half the movement, so you can use it as a depth control for the whole slot.

[Scene 8] Turn on Sync and the Rate knob becomes a Division menu, locked to your project tempo. Eighth notes on the cutoff gives rhythmic movement. Bipolar on swings above and below the knob; off, it only goes one way. If you are working in a DAW the synced LFO follows the project tempo. In the standalone app, where there is no host tempo, it assumes one hundred and twenty beats per minute.

[Scene 9] Phase sets where the LFO starts at each note. Level sets its depth. LFOs are per voice, so every note you play starts its own cycle. Because each note has its own LFO, chords can sound alive and slightly different on each note, unless the phase is set to start them all together.

[Scene 10] One rule to remember: voice modulators, envelopes, LFOs, performers, velocity, key track and random, cannot control the master effects. Try it and you get a message. Effects are shared by all notes, so for them use a macro, mod wheel or aftertouch. That is why macros, the mod wheel and aftertouch are the way to move effects. They are the same for every note, so they are meaningful for the shared effects at the end of the chain.

[Scene 11] Drag the ModWhl chip instead and it works. Move the wheel on the on-screen keyboard and the reverb opens.

[Scene 12] To remove a modulation, right-click the knob and choose Remove. Undo and Redo cover assignments, so you can experiment freely. You can have up to two hundred fifty-six assignments, several sources on one knob, and one source on many knobs, so do not be afraid to layer modulation.

[Scene 13] Envelopes for the shape of a note, LFOs for repeating movement. In video six we meet macros, the controls that rename themselves.

> **TIP.** Pause for about one second between scenes and let on-screen actions finish before the next sentence starts.